



Separating project risk from the time value of money: A step toward integration of risk management and valuation of infrastructure investments

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Abstract

The rationale for using heuristics to establish a risk premium that is added to the risk-free rate to obtain the value of an investment is questioned and an alternative method, termed decoupled net present value (DNPV), is proposed. Rather than using utility theory concepts to decrease the value of uncertain cash flows, the risks associated with project cash flows are discretely quantified using insurance and contingent claim valuation concepts. Synthetic insurance premiums are designed to “protect” the value of expected cash flows which are treated as additional project costs. Because identified project risks are quantified in financial terms and treated as a real cost to the project, DNPV allows business executives to evaluate the effect on the value of the project of different risks and select management techniques that are deemed more effective. Hence, DNPV is both a valuation methodology and a risk management tool.

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1. Introduction

Despite all the work done on investment valuation, there is not a single valuation methodology that is used consistently across different industries. The most popular method to evaluate an investment opportunity such as an infrastructure project is to forecast the project cash flows and discount them by a risk-adjusted discount rate (α) to calculate the project net present value (NPV) or internal rate of return (IRR). A positive NPV (or $IRR \geq \alpha$) is considered a potentially attractive investment opportunity whereas a negative NPV (or $IRR < \alpha$) is considered unattractive. As its name implies, the risk-adjusted discount rate accounts for the project risk by simply adding a risk premium (r_p) to the risk-free rate (r), that is $\alpha = r + r_p$. The general idea behind the use of a risk-adjusted discount rate is that investors need to be compensated in two ways: time value of money and the project risk. The time

value of money is represented by the risk-free rate and compensates investors for parking their money in a secure investment that yields a known amount over a period of time. The project risk is represented by the risk premium which is used to estimate the compensation above the risk-free interest rate that a rational (risk-averse) investor would demand for taking on additional risk rather than investing in government treasury bonds. However, lumping risk with the time value of money immediately creates two problems: (i) it disconnects project risks from their actual source (i.e., cash flows); and (ii) it implicitly assumes that risk and time are interchangeable parameters. This apparently innocuous connection, now widely accepted by practitioners, causes two distinct problems: (i) On the valuation side: NPV techniques understates the value of future (positive or negative) cash flows, particularly for long-term investments such as infrastructure projects that have large initial capital outlays and potentially long operational (payback) periods; and (ii) On the management side: NPV techniques makes it extremely difficult to assess the effectiveness of risk management measures to mitigate some of the project risks.

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If future cash flows are positive, the NPV method can grossly undervalue an investment opportunity; more onerously, the opposite is true if future cash flows are negative. Unfortunately, rather than amending the problem (i.e., separating risk and time value of money) and assigning risk to where it belongs (i.e., to the uncertain cash flows), most efforts have been devoted to fine-tuning the discounting method such as the capital asset pricing model (CAPM) to account for the project specific risks (e.g., Butler and Pinkerton, 2006; Butler et al., 2011; Myers and Turnbull, 1977), or to correct (mostly increase) the NPV results by the inclusion of the value of real options embedded in projects (e.g., Cheah and Liu, 2006; Chen et al., 2009; Copeland and Keenan, 1998; Garvin and Cheah, 2004; Liu et al., 2014; Myers, 1984; van Putten and MacMillan, 2004). On the other hand, although good efforts have been made to account for the variability of cash flows, particularly for infrastructure investments, using Monte Carlo simulations (e.g., Chiara and Garvin, 2008; Girmscheid, 2009; Liou and Huang, 2008; Ye and Tiong, 2000, 2003), in most cases, risk is double counted as, in theory, risk is already considered in the choice of the risk adjusted discount rate. In addition, these methods are mathematically complex and difficult to implement in standard valuation tools. More importantly, it is extremely difficult to convey any results in terms that are familiar to decision makers.

NPV (or IRR) users who want to use the discount rate as a proxy for risk generally have one main question: What is the appropriate risk-adjusted discount rate that should be used to estimate the project NPV and, hence, the value of the investment opportunity? Understandably, because risk and time value of money are separate variables (Robichek and Myers, 1966), a unique and consistent answer cannot be provided, as evidenced by guidelines typically found in numerous capital budgeting articles (e.g., Davies et al., 2012; McDonald, 1997) and investment textbooks (e.g., Trigeorgis, 1999: 39; Campbell et al., 2001:136). As a result, each organization establishes its own set of recommended rates (i.e., hurdle rates) to compare potential projects to in order to assess their financial viability. For instance, some authors promote the use of CAPM to estimate the hurdle rate of the project (e.g., Fama, 1977). CAPM dictates that a risk premium should only be added for taking non-diversifiable risk (i.e., market risks); that is, investors should not be compensated for assuming diversifiable risks because this has already been incorporated into the investment selection (Sharpe, 1964). Hence, an asset (or investment) is correctly priced when future cash flows of the asset are discounted at a rate suggested by CAPM. Extensions of CAPM to value the return on investment on real projects can be found in the literature (e.g., Fama, 1977; Myers and Turnbull, 1977). However, Bhattacharya and Leach (1999) demonstrated that a historical proxy for the discount rate cannot be used even for expansion projects that supposedly should have the same risk profile of the company.

Although CAPM may be a valid model to estimate the return of stocks, bonds, or other liquid financial instruments that can be bought and sold rather easily to create a well-diversified portfolio to the point that non-systematic risks are eliminated (i.e., averaged out), when investing in non-financial (illiquid) assets such as infrastructure investments, such diversification

cannot be attained. Rational investors must consider both theoretically diversifiable (i.e., private) and non-diversifiable (market) risks associated with the revenues and the expenditures of the potential investment opportunity. In practice, this means that investors hope to select hurdle rates that account for all risks associated with the project not just market risks. Unfortunately, selection of this “perfect” hurdle rate is further obscured by the manner in which different risks are accounted for. For instance, in the drug industry, to account for the probability of success of new drugs, cash flows from potential new products are adjusted by the estimated probability of success and the adjusted cash flows are then discounted using the industry hurdle rate. This procedure is termed the risk-adjusted NPV or expected NPV (Steward et al., 2001). Similarly, in the construction industry, contractors add contingency budgets to account for project private risks (Baccarini, 2004) and discount future cash flows using an industry-specified hurdle rate. In other industries, private risks are accounted for in the same manner as market risks, that is, by adding a risk premium to the risk free rate. For instance, some business appraisers calculate risk premiums using the Total Beta concept (e.g., Helfenstein, 2009). Others researchers, motivated by the difficulties encountered when evaluating long term public investment projects using constant discount rates, have proposed a decreasing discount rate for longer term projects (e.g., Grollier, 2002). As a result of these often conflicting ideas, investors resort to heuristic arguments to account for the identified risks (systematic and non-systematic) and select the “appropriate” risk premium to estimate the project NPV. Consequently, the selection of an appropriate discount rate that consistently reflects the risk associated with real projects and business valuation¹ remains a guessing exercise at best because correlating identified project risks to a single factor (i.e., the risk premium) using heuristics alone simply cannot be as straightforward as is often hoped. At worst, this may not even be a valid approach, as discussed next.

The difficulties of consistently correlating a risk premium to project risks by lumping the time value of money and risk together in a single number, α , and correlating it to the return on a real investment were examined by Robichek and Myers (1966), who noted that cash flows are the unknown quantities (i.e., uncertain), not the discount rate, and advocated for the use of certainty equivalent (CE) cash flows. Hamada (1977) pointed out that the extension of risk-adjusted discount rates to the valuation of projects with cash flows that do not follow random walk processes can be very dangerous and exhorted the finance profession to use the CE methods. Bhattacharya (1978) showed that although the accuracy of the risk adjusted discount method is good for cash flows that replicate exponentially, the accuracy of the methodology decreases for cash flows that follow mean-reverting processes. Based on CE concepts, Halliwell (2001) listed six inconsistencies associated with the risk-adjusted method and, more recently, (Halliwell, 2011) demonstrated that risk adjusted discount rates are

¹ The lively discussion in the area of business appraisal regarding the use of Total Beta instead of Beta to capture company specific risks (e.g., Butler et al., 2011) when estimating risk premiums provides testimony of the confusion created by mixing risk and time value of money.

conditionally valid for cash flows that increase exponentially with time. Zeckhauser and Viscusi (2008) noted that the preferred approach to address risk should be to convert risky cash flows to CE cash flows and then perform the discounting using risk-free rates. In summary, although the limitations associated with the use of a risk-adjusted discount rate to estimate the value of a project have been widely recognized for over 40 years, and numerous attempts have been made to correct its shortcomings, because of their simplicity, risk-adjusted discount rate methods continue to be the most widely used valuation methodology employed by corporations and investors alike (Block, 2007; Ryan and Ryan, 2002; Triantis, 2005).

The main problem with adding a risk premium to the risk-free rate to account for project risk is that the application of NPV may lead to significant under investment (or favoring less profitable short-term investments over more profitable long-term ones) and/or selecting projects with understated future potential liabilities (Espinoza and Morris, 2013), thus making it difficult for business executives to consistently make sound management decisions. Savvy investors and experienced managers understand the NPV method significantly understates the value of an investment (and/or future liabilities) and rely on different metrics, heuristics and gut feeling to accept/reject projects despite what NPV suggests. However, in an environment of tight corporate governance, discarding the results of established NPV protocols to favor experience and gut feeling may not be always a valid alternative. As a result, the growth of a firm could be impacted by the application of NPV methods particularly when the discount rate is used as a proxy for risk.

2. Decoupled Net Present Value

In this paper, a simple procedure that uses the probability distribution of the cash flows (either actual or risk neutral probabilities) is proposed to mimic a stream of riskless cash flows which are then discounted using a risk free rate to obtain the value of the investment opportunity. The proposed procedure, termed decoupled net present value (DNPV), effectively decouples risk from the time value of money (i.e., de-risks the cash flows). The DNPV method can be viewed as an extension of the CE method. Although the CE method, that follows directly from utility theory concepts, has been considered a more robust, flexible, and powerful tool than conventional NPV methods (Hamada, 1977), it has seldom been used in practice because of the difficulties associated with the application of utility theory to estimate the cash flow reduction parameters that would render the cash flows riskless. In the proposed procedure, DNPV uses insurance and contingency claim valuation concepts to estimate synthetic insurance premiums which are then subtracted from the stochastic cash flows to render a synthetically riskless cash flow. For the cases where price variability can be observed (e.g., commodity prices), insurance premiums are equivalent to financial put options.

In what follows, the decoupled present value (DPV) of the revenues collected over n periods is defined as the sum of the synthetically riskless revenues (i.e., expected value of the revenues

$\tilde{V}_t$ minus the cost of the synthetic insurance premiums) discounted using the risk-free rate (r) as follows:

$$DPV(V, r, R_V) = \sum_{t=0}^n \frac{\tilde{V}_t - \tilde{R}_{Vt}}{(1+r)^t} \quad (1)$$

$$\tilde{R}_{Vt} = \sum_{j=1}^{m_t} \tilde{R}_{Vt}^j \quad (2)$$

where $\tilde{R}_{Vt}^j$ is the synthetic insurance premium associated with risk j and $\tilde{R}_{Vt}$ represents the total synthetic insurance premium for the revenues at time t , and m_t is the number of risks at time t . As discussed above, Eq. (1) is similar to the expression presented by Robichek and Myers (1966) with CE factor equal to $1 - \tilde{R}_{Vt}/\tilde{V}_t$. Also, the form of Eq. (1) is similar to the equation derived by Fama (1977) for valuing risky financial assets taking into consideration only market risk. Similarly, the DPV of the expenditures accrued over n periods can be defined as the sum of the synthetically riskless expenditures (i.e., expected value of the expenditures $\tilde{I}_t$ plus the cost of the synthetic insurance premium $\tilde{R}_{It}$) discounted using the risk-free rate as follows:

$$DPV(I, r, R_I) = \sum_{t=0}^n \frac{\tilde{I}_t + \tilde{R}_{It}}{(1+r)^t} \quad (3)$$

$$\tilde{R}_{It} = \sum_{j=1}^{m_t} \tilde{R}_{It}^j \quad (4)$$

where $\tilde{R}_{It}^j$ is the synthetic insurance premium associated with the expenditure risk j and $\tilde{R}_{It}$ represents the total synthetic insurance premium for the expenditures risk at time t . Note that the synthetic insurance premium for the revenues reduces the value of the revenues whereas the synthetic insurance premium for the expenditures increases the total investment cost. The project DNPV can be calculated in terms of the DPV of the revenues and expenditures as follows:

$$DNPV(V, I, r, R) = DPV(V, r, R_V) - DPV(I, r, R_I) \quad (5)$$

It follows that:

$$DNPV = \sum_{t=0}^n \frac{\tilde{V}_t - \tilde{I}_t - \tilde{R}_t}{(1+r)^t} \quad (6)$$

where $\tilde{R}_t = \tilde{R}_{Vt} + \tilde{R}_{It}$ represents the total expected synthetic insurance premium at time t . From Eq. (6), it follows that DNPV can be expressed as the difference between the present value of the expected revenues and the expected expenditures minus the present values of the total expected synthetic insurance premiums as follows:

$$DNPV = PV(\tilde{V}, r) - PV(\tilde{I}, r) - PV(\tilde{R}, r) \quad (7)$$

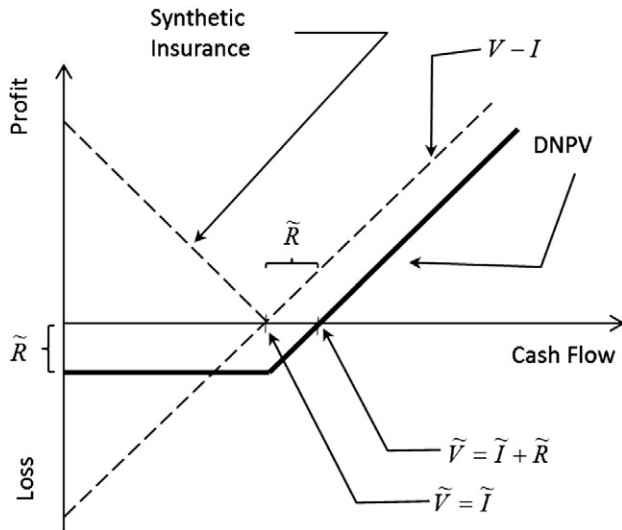


Fig. 1. Representation of DNPV.

If actual insurance products and/or financial instruments (e.g., put options) were available in the market to represent the synthetic insurance premiums, Eq. (7) could be represented by Fig. 1. As shown in this figure, the profit–loss pattern of the proposed DNPV is similar to that of a protective put (or a long call option position) of an asset with an expected value $\tilde{V}$, an expected exercise price $\tilde{I}$, and option premium equal to $\tilde{R}$. Therefore, everything else being equal (i.e., expected revenues and expenditures), Fig. 1 shows that riskier projects (characterized by higher premiums) will result in lower project valuations. In other words, DNPV illustrates how a large upside potential is counterbalanced by an equal and opposite downside potential and, it is the latter that risk-averse investors worry the most as a series of “unlucky” events could bankrupt them. An investor having a long call option position can easily lose its premium if the stock ends below the exercise price. However, Different from financial options where the losses on long call option positions are limited to the premium $\tilde{R}$, in a real investment, losses can be greater than the option premium as the conditions that made the option valuable could change between the time the initial investment is made and revenues are generated. Nevertheless, the analogy is a powerful one and it is further enhanced by the fact that, in financial options, as DNPV, the stock price (i.e., revenues) and the exercise price (i.e., the expenditures) are discounted back using risk free interest rates.

Using a financial options analogy, a DNPV rule (i.e., accept the project if $\text{DNPV} > 0$) would be equivalent to investing only if the project is “in the money” by at least the value of the premium (i.e., the expected revenues are at least equal to the expected expenditures plus the risk premium). This simple analogy that provides a link between the proposed methodology and options analysis allows investors to use well-known risk management strategies to hedge portions of the risks they feel they are not best to bear (Pergler and Freeman, 2008). From the project risk profile defined to estimate the synthetic insurance premiums, the investor/manager could easily identify potential future actions along with its probability of occurrence and add it

to the initial estimated DNPV. Thus, Eq. (7) could be expanded as:

$$\text{DNPV}_T = \text{DNPV}_0 + \sum_{i=1}^n \text{DNPV}_i \quad (8)$$

where DNPV_T represents total project DNPV enhanced by all identified options associated with the project (i.e., DNPV_i) and added to the initial DNPV_0 . The advantages of the proposed expression over real option valuation (ROV) where NPV is enhanced by the options embedded in the project (e.g., Trigeorgis, 1999; van Putten and MacMillan, 2004) are several: (i) DNPV is consistent with financial options analysis in the use of the risk free discount rate to discount cash flows; (ii) like its financial counterpart, risk premiums (i.e., cost of risk) are based on the underlying project risk profiles; (iii) DNPV avoids the use of an arbitrary risk adjusted discount rate that may not be consistent with the project risk profile; and (iv) the profit–loss profile of DNPV is consistent with that of a long call position; and (v) it acknowledges the potential for realizing investment losses (i.e., DNPV is not always greater than zero). In summary, the proposed DNPV methodology is not only intuitive but also complies with the seven criteria for a viable valuation model proposed by Babbal and Merrill (1998), that is: (i) it is meaningful; (ii) it is implementable; (iii) it is consistent; (iv) it can be calibrated; (v) it is based on non-controversial principles; (vi) its parameters can be specified and audited; and (vii) it is additive.

Hence, to estimate the project DNPV, all identified project risks (systematic and non-systematic) must be priced. By identifying and quantifying each project risk, these risks can be managed using well known risk management techniques including: (i) risk avoidance (e.g., withdrawing from the investment opportunity); (ii) risk reduction (i.e., reducing the probability of revenue reductions or cost increases occurring by implementing preventive maintenance, audits and compliance programs); (iii) risk mitigation (i.e., reducing the severity of a risk event if it were to take place); (iv) risk transfer (e.g., contract terms, future contracts, insurance, outsourcing); and (v) risk retention (i.e., adsorbing the risk that the investor understands by self-insurance). This represents a significant step forward in risk management as the proposed DNPV method provides managers with a direct link between cause and effect (i.e., risk/reward) of their investment decisions. Hence, the proposed DNPV method not only provides a consistent investment valuation methodology that is in line with the CE method but also provides a tool for integrating risk management with risk quantification. The main difference with the CE method is that rather than selecting a single factor to reduce the cash flows on a given period, the DNPV method promotes the identification of potential risks that could affect the project cash flows and estimates the potential cash flow reduction associated with this risk using insurance and/or contingent claim valuation concepts.

3. Cost of risk: synthetic insurance premiums

In a certain (i.e., risk free) environment with known future revenues (V_i), expenditures (I_i), and, therefore, cash flows (C_i =

$V_t - I_t$), the NPV methodology is a valid approach that allows investors to compare different investment opportunities. As discussed further in a subsequent section, even for uncertain cash flows, NPV is a useful tool if the discount rate is not used as a proxy for risk and it only reflects the investor's cost of capital. In most real investment and capital investment projects, investors/managers need to forecast the expected revenues ($\tilde{V}_t$) and expected expenditures ($\tilde{I}_t$) to estimate the expected cash flows ($\tilde{C}_t$). Hence, business executives must weigh the risk of obtaining lower-than-forecasted cash flows through the life of the project due to revenues being lower than forecasted ($V_t < \tilde{V}_t$) and/or capital/operational expenditures being higher than forecasted ($I_t > \tilde{I}_t$). Thus, project risks can be viewed as the amount of possible negative deviation between the forecasted and actual future cash flows (i.e., lower than the expected cash flows, $C_t \leq \tilde{C}_t$) and these negative deviations can be due to market risks as well as private risks. In other words, the cash flow shortfall below the forecasted cash flows can be viewed as a potential loss, and the higher the potential shortfall, the higher the potential risk. Rational investors expect to be compensated for providing capital for risky projects and, naturally, the higher the total project risk, the higher the compensation they expect.

Rather than accounting for the risky cash flows by adding a premium to the risk free rate, Robichek and Myers (1966) proposed an alternative procedure (i.e., the CE method) that reduces the risky cash flows to a level that would make investors indifferent to receiving either the risky expected cash flows ($\tilde{C}_t = \tilde{V}_t - \tilde{I}_t$) or the reduced (i.e., riskless) cash flows, which could be discounted using the risk free rate to determine the value of the project. As indicated above, the CE method is considered a more robust methodology because by separating the effect of time value of money and cash flow risk, it eliminates all the inconsistencies associated with adjusting the risk free rate to account for project risk. However, the application of CE methods requires the use of utility theory concepts (not widely used in practice) to determine the appropriate reduction factors needed to render the cash flows riskless. Interestingly, business executives are not uncomfortable with using information from experience professionals (e.g., operation managers, marketing managers; engineers) to estimate expected ranges of cash flows. In some instances, data may already be available to facilitate the description of the possible range of values that cash flows could take over the periods analyzed. Therefore, probability density functions (PDFs) can be developed to describe the project cash flows and the different risks associated with them. These PDFs can then be used along with the proposed procedure to account for the project risks, and render the projected cash flows synthetically riskless (i.e., de-risk the cash flows). For instance, for investments on conventional hydropower plants, one could easily identify the following risks associated with the expected revenues: electricity price fluctuations, variability in the amount of water available to produce energy, partial or total disruptions of the plant operations. A similar list could be developed on the expenditure side.

Similar to the CE method, an investor would be indifferent between the expected cash flows and the riskless cash flows if there were a synthetic insurance premium (essentially a put option) that would “guarantee” the expected cash flows. Because the

expected cash flows are reduced by the synthetic insurance premiums, in accordance with the CE method, the resulting riskless cash flows can be then discounted using the risk free rate. Insurance premiums can be derived from concepts used in the insurance industry, and in some cases (e.g., traded commodities), insurance premiums can be estimated based on no-arbitrage (i.e., option pricing) principles (Embretchs, 2000). Hence, if cash flows can be estimated and the corresponding risks identified, then risk premiums associated with the uncertain cash flows could be estimated by applying insurance and/or real option concepts. Some of the elements discussed in this paper are discussed by Carmichael et al. (2011) who developed a methodology to estimate the upside value of a real option using standard NPV analysis. Although the methodology has the same limitations identified in most ROV (e.g., the value of the asset without the option is estimated assuming a risk adjusted discount rate), the general concept is valid (i.e. assigning a PDF to the cash flows and capturing the upside potential) and it can be corrected by discounting the cash flows using risk-free rates as proposed in this paper. The corrected approach can then be extended to estimate the downside potential (i.e., the insurance premium) of the investment.

In the DNPV methodology, investors are viewed as insurance/hedge providers of the project forecasted cash flows with the estimated synthetic insurance premiums as their compensation for the risk assumed. Because these types of investments are not liquid, the price of risk (i.e., insurance premiums) will depend upon the investor's comfort/knowledge of the potential project risks and their ability to remain solvent during periods of adverse outcomes as would be the case for insurance providers. In what follows, it is assumed that all project risks that could result in a reduction of the cash flows below the expected values are accounted for by synthetic insurance premiums that are treated as costs to the project. For insurable risks² or risks that can be hedged (e.g., financial risks), premiums could be actual costs to the project. For uninsurable risks (or risks that cannot be hedged), synthetic³ insurance premiums will represent the investor's expected compensation for taking on the project risks. Although for some project risks (e.g., fire, flood), there are products available in the insurance market, insurance products will not be available for most project risks. However, by thinking in terms of insurance premiums and hedge products, albeit synthetic, investors are forced to think of all the possible risks that could affect the project cash flows and how these risks could impact the financial outcome of an investment. Furthermore, because the DNPV method applies to both non-systematic and systematic risks, the cost of risk (i.e., synthetic insurance premiums) can be evaluated in a consistent manner throughout the life of the project.

² In some cases, even if insurance products are available, a business may consider self-insurance to be more cost-efficient as many large corporation have done with the creation of captive insurance companies.

³ The word synthetic is used to indicate that the insurance product may not be commercially available, in which case it is created to replicate the financial effect of an insurance product. If it exists (e.g., put options for commodity prices), market information can be used to assess the cost of risk.

Considering risk as a cost that affects cash flows is a more natural progression of a well-established business practice of buying insurance products to protect against certain insurable risks. Initially, these synthetic insurance products could be estimated subjectively in cases where data is not readily available or from available empirical data in cases where data has been collected and appropriately processed. When data is available, PDFs of the main parameters or more sophisticated techniques, such as option pricing methods, could be used (Espinoza, 2011). In fact, because many investors/managers are already familiar with the techniques for assessing risk and developing risk profiles for different project risks, this practice can be seamlessly integrated into existing project valuation techniques. Hence, estimating synthetic insurance premiums should not be difficult to implement.

If project risks are viewed as synthetic insurance premiums (or put options) that, if purchased, could reduce the project's future expected cash flows, then, in line with the CE method, a risk premium added to the risk free rate is no longer necessary as the project risks have already been considered in the estimation of the synthetic riskless cash flows. Synthetic insurance premiums that account for losses in expected cash flows due to lower revenues ($\tilde{R}_{Vt}$) and/or higher expenditures ($\tilde{R}_{It}$) at time t could be estimated for each identified risk and subtracted from the project cash flows. Hence, the quantities $(\tilde{V}_t - \tilde{R}_{Vt})$ and $(\tilde{I}_t + \tilde{R}_{It})$ can be considered as the synthetic riskless revenues (or the certainty equivalent revenues) and synthetic riskless expenditures (or the certainty equivalent expenditures), respectively or more generally, $(\tilde{V}_t - \tilde{I}_t - \tilde{R}_t)$ can be considered as the synthetic riskless cash flow.

Assuming that the cash flows are random variables with expected values at time t equal to $\tilde{C}_t$, then the synthetic insurance risk premium ($\tilde{R}_t$) to cover the potential loss of cash flow (L_t) due to a reduction of the expected revenues and/or increase in expected expenditures can be defined as (Embretchs, 2000):

$$u[\tilde{C}_t - \tilde{R}_t] = E[u(\tilde{C}_t - L_t)] \quad (9)$$

where u is the investor's utility function for a given risk⁴ (i.e., $u' > 0$ and $u'' \leq 0$) and $E[\bullet]$ is the expectation operator. From Jensen's inequality, the concavity of u requires that:

$$\tilde{R}_t \geq E[L_t] = \tilde{L}_t \quad (10)$$

The most common representation of an insurance premium is given by the Standard Deviation Principle (Embretchs, 2000):

$$\tilde{R}_t = \tilde{L}_t + \lambda \tilde{L}_t = (1 + \lambda c_v) \tilde{L}_t \quad (11)$$

For the case of a linear utility function (i.e., $u'' = 0$ or risk neutral), the risk premium is equal to the expected value of the loss (i.e., $\lambda = 0$). If the PDFs of the main components of the cash flows (i.e., revenues and expenditures) have been defined, then the PDF of the potential cash flow losses (L_{Vt}) due to a reduction

of the random revenues (V_t) below the expected revenues ($\tilde{V}_t$) is given by:

$$L_{Vt} = (\tilde{V}_t - \tilde{V}_t^-) \times \Pr[\tilde{V}_t > V_t] \quad (12)$$

where $\tilde{V}_t^-$ represents the expected value of the revenue downside. Similarly, the PDF of the potential cash flow losses (L_{It}) due to an increase of the random expenditures (I_t) that are above the expected expenditures ($\tilde{I}_t$) can be expressed as follows:

$$L_{It} = (\tilde{I}_t^+ - \tilde{I}_t) \times \Pr[I_t > \tilde{I}_t] \quad (13)$$

where $\tilde{I}_t^+$ represents the expected value of the expenditure downside (i.e., expenditures greater than anticipated). Hence, Eqs. (11), (12) and (13) can be used to capture the downside and estimate the synthetic insurance premium for the project revenues and expenditures, respectively.

For market risks, stochastic processes and risk neutral probabilities can be used to model the variability of the revenues reasonably well. In such cases, put option pricing techniques could be used to price the value of synthetic insurance premiums.⁵ If the asset under creation exists in the market (e.g., the price of oil for an oil exploration project), investors could use quoted prices to estimate the synthetic insurance and decide whether or not to hedge against that risk. In general, Eqs. (12) and (13) are equivalent to put option pricing formulae because they represent a truncated distribution derived from an underlying variable (i.e., the project revenues and/or expenditures). Furthermore, for some particular cases, closed form solutions for option pricing can be used to estimate the synthetic insurance premiums due to technical risk (e.g., Espinoza, 2011).

4. Implied risk adjusted discount rate

The power of the proposed DNPV methodology is perhaps best demonstrated by estimating, rather than assuming, an implied risk-adjusted discount rate to calculate the project NPV. An implied risk adjusted discount rate (α) can be found after the DNPV has been calculated using the cost of risk as defined above. The implied risk adjusted discount rate represents the average discount rate to be used to discount the expected cash flows for the specified period and results in a NPV that equates to DNPV, that is:

$$\text{NPV}(\tilde{V}, \tilde{I}, \alpha) = \text{DNPV}(\tilde{V}, \tilde{I}, r, \tilde{R}) \quad (14)$$

Although the value of the implied α calculated using Eq. (14) can be viewed as a proxy for the project risk, this value differs from the prescribed ones typically found in the literature. In this article, risks are treated as real costs to the project which vary with time, so to replicate the NPV evolution with time, a variable discount rate would be required. However, the implied⁶ α could

⁵ Option pricing models do not include the expected return of the underlying asset and, therefore, do not depend on the investor's risk preferences.

⁶ The word implied is used to emphasize that the risk adjusted discount rate has been back-calculated from the DNPV methodology and it is a function of the project duration and risk profiles.

⁴ Different utility functions apply for different risks.

be used to represent an average risk adjusted discount rate for the project.

Of course, once the DNPV is calculated, there is no need to estimate the implied risk adjusted discount rate; it suffices to know whether or not $DNPV \geq 0$. However, since this parameter is ubiquitous with decision makers, it can be easily related to other familiar financial parameters such as IRR and WACC and it can be used as a proxy for the project risk for the proposed duration. It is noted that the implied risk adjusted discount rate is an average value that overvalues the expected returns at some periods but undervalues them at others such that net effects are cancelled out. It is also noted that the implied risk adjusted discount rate should not be used to calculate the NPV for a different period even if the project riskiness remains constant throughout. Moreover, it will be shown in the next section that, depending upon the risk profile, the implied risk adjusted discount rate depends upon the length of the project and, hence, it cannot be assumed constant even for the same risk profile. This is in agreement with the findings reported by [Bhattacharya \(1978\)](#) for traded securities obtained assuming mean-reverting cash flows.

5. Financial and risk performance

In this paper, it is contended that the use of NPV with a specified cost of capital (e.g., WACC if the project is financed with a mix of debt and equity) and DNPV to discount the same expected cash flows using risk free rates and accounting for risk by the proposed synthetic insurance premiums are not incompatible concepts, they simply measure two different metrics: financial performance and risk performance. Unfortunately, the long-term practice of mixing the time value of money and risks have led to a significant confusion regarding the meaning of expected return (i.e., WACC) and project risk.

A financial metric should measure the ability of a project/business to compensate its equity and debt holders with their expected returns. To that end, the expected cash flows need to be discounted using the investor's WACC. The cost of equity corresponds to the average for the industry sector to which the project belongs and can be estimated using publicly available market data. For an investor to park funds on a given (illiquid) project/business, the expected return should be at least equal to the market return for the given industry otherwise they would be better off investing in the industry as a whole through market (liquid) securities. However, WACC does not (and should not) consider the project specific risks. In other words, WACC is exogenously imposed to the expected cash flows. In summary, a project/business considered to be financially feasible should have a $NPV \geq 0$ (or equivalently $IRR/WACC \geq 1$).

To evaluate the project risk performance, the proposed DNPV methodology along with the synthetic insurance premiums (i.e., cost of risk) detailed above should be used. To compare with IRR and/or WACC, the project implied risk adjusted discount rate (α) could be estimated using Eq. (14). Because the calculated cost of risks is project specific, DNPV (and therefore α) is endogenous to the project. A project/business is considered to be feasible from

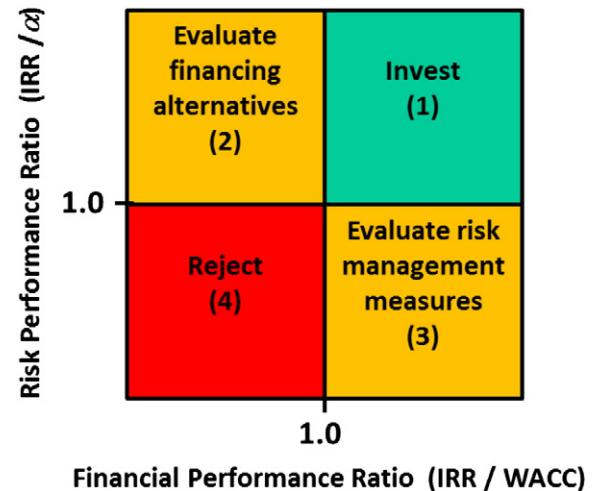


Fig. 2. Risk and financial performance chart.

the risk performance standpoint if $DNPV \geq 0$, or equivalently, if $IRR \geq \alpha$ (i.e., $IRR/\alpha \geq 1$).

To determine the feasibility of a project/business, both the financial performance (NPV) and risk performance (DNPV) should be evaluated. Hence, an investment is said to be feasible if both the financial and risk performance measures (NPV and DNPV, respectively) are greater than zero (i.e., fall within Box 1 in Fig. 2). DNPV introduces a new dimension in the evaluation of project feasibility (Boxes 1 and 2 in Fig. 2). The proposed metric can be used to evaluate competing projects for a limited amount of funding. Depending upon where the evaluation of a project plots in the risk performance–financial performance chart, an investment may require financial restructuring (if it falls in Box 2) or modification to the risk mitigation measures (if it falls in Box 3) in order to become feasible. If more than one project falls in Box 1, it is recommended that the project with the highest DNPV (i.e., highest IRR/α) should be selected. The ability of DNPV to quantify the effect of risk management measures makes the proposed metric a powerful risk management technique.

6. Simplified application examples

6.1. Introduction

In what follows, the synthetic risk premiums for three simplified cases (i.e., single source of revenues) that will be used as building blocks for more realistic financial analysis are subsequently presented, namely: (i) Case 1: Loss of revenue due to a permanent shutdown (i.e., the project may no longer generate revenues at some point during the contract period); (ii) Case 2: Loss of revenue due to a temporary shutdown (i.e., the project may not generate revenues temporarily in a given period at some point during the contract period⁷); (iii) Case 3: Reduction of revenues due to changes in market prices (e.g., commodities). For illustrative purposes the implied risk adjusted discount rates that

⁷ Temporary project shutdown could also be applied to account for schedule delays that could result in the loss of revenues for a period of time.

are consistent with the project risk profiles (i.e., risk premiums) are calculated for each of these cases. This is a significant departure from the current practice that assumes a risk adjusted discount rate to be used to calculate the project NPV. Assuming that the revenue risk profile is known (i.e., the cost of risk can be estimated), the project DNPV can be written as:

$$\text{DNPV} = \sum_{i=1}^n \frac{(\tilde{V}_i - R_i)}{(1+r)^i} - \tilde{M}_o = 0 \quad (15)$$

where $\tilde{V}_i$ is the project expected risky revenues and R_i is the synthetic insurance premium that depends on the project risk profile, $\tilde{M}_o$ is the value of the investment, and n is the revenue period (e.g., lease agreement, patent duration).

6.2. Case 1: loss of revenue due to a permanent shutdown

To facilitate the discussion, it is assumed that the stream of revenues is constant and the probability of complete project shutdown (due to technical, non-systematic factors) is also constant with time (i.e., at any time, there is an equal probability of the project shutting down forever, resembling the case of Tokyo Electric Power's Fukushima Daiichis nuclear power plant in March 2011) and equal to ρ . Once the project is shutdown no additional revenues will be collected. Since the value of the project at each period depends on the future stream of revenues, the revenues at risk decreases with time because the value of the project decreases from $\tilde{M}_o$ (i.e., the initial project value) to zero at the end of the contract period (assuming no terminal or salvage value). As illustrated below, at any time during the contract period, there are two possible events: either the project is permanently shut down or else it continues to generate a stream of revenues. For this simple binary probability distribution, the risk profile for this project is illustrated in Table 1 where the remaining expected value of the project at any future time i is represented by $\tilde{M}_i$ (i.e., the summation of Eq. (15) from $j = i$ to n).

Using the concepts described above, to use the DNPV method, the cost of a synthetic insurance premium that would protect the investor against obtaining less than the expected value of the investment at a given time ($\tilde{M}_i = M_i \times (1-\rho)$) needs to be calculated. Using the project risk profile shown in Table 1 (3rd column), the corresponding synthetic insurance risk profile that describes the potential loss distribution can be constructed as shown in the 4th column of Table 1. From Table 1, it follows that the expected value of the loss ($\tilde{L}_i$) is $\rho \times \tilde{M}_i$, the standard deviation (σ_v) is $\sqrt{\rho \times (1-\rho)} \times \tilde{M}_i$, and coefficient of variation (c_v) is $\sqrt{\rho/(1-\rho)}$. Assuming that the investor uses the standard deviation principle as an appropriate measure of the synthetic insurance premium,⁸ then the synthetic insurance premium (R_i) for this case can be written as:

$$R_i = \tilde{L}_i(1 + \lambda c_v) = \Theta \tilde{M}_i \quad (16)$$

Table 1

Project risk profile at any period (complete shutdown).

Event (E)	Probability, $\text{Pr}(E)$	Value of the project	Insurance payout
Shutdown	ρ	0	$\tilde{M}_i$
No shutdown	$1 - \rho$	M_i	0

where:

$$\Theta = \rho \times (1 + \lambda c_v) \quad (17)$$

and λ is a loading factor greater than zero.⁹ Eq. (16) represents the synthetic insurance premium designed to account for the risk of obtaining less than the expected value borne by the investor. Because the cost of this synthetic insurance premium is proportional to the remaining value of the project, the risk decreases with time as $\tilde{M}_i$ decreases. For this investment example, the cash flow model is as shown in Table 2. Because the risk associated with not getting the expected revenues in any given period is considered by the synthetic insurance premium, Eq. (16), and there are no other risks associated with the project revenues, the value of the project can be estimated by replacing the values from Table 2 in Eq. (15), yielding:

$$M_o = V - F(r + \Theta) \quad (18)$$

where F is the inverse of the capital recovery factor given by Eq. (A.4) in Appendix A. For this particular case, the implied risk-adjusted discount rate to account for the probability of permanent project shutdown at any time during the contract period is given by Eq. (19):

$$\alpha = r + \Theta \quad (19)$$

This equation indicates that for the selected project risk profile, the appropriate equivalent risk adjusted discount rate (α) to be used to discount the project cash flows is independent of the length of the project. Note that the cash flows are estimated based on expected values but the value of the project calculated in Eq. (18) is based on the actual value that would be paid by an investor after considering the project risk. As such, Eq. (18) is the general case of a similar expression derived by Ruback (2011) for cash flows that are biased measures of expected cash flows. Defining the risk factor for the revenues (η) as the ratio of the synthetic insurance premium to the expected revenues, the risk factor would be:

$$\eta_i = \left(\frac{1}{r + \Theta} - (1 + r + \Theta)^{i-1} \left[\frac{1}{r + \Theta} - F(r + \Theta) \right] \right) \Theta \quad (20)$$

It follows from Eq. (20) that, as expected, the risk factor decreases monotonically with time indicating that every year that passes without permanent project shutdown puts less of the initial investment at risk. Assuming $\rho = 20\%$ (i.e., $c_v = 0.5$), $r = 3\%$, and $n = 10$, the appropriate implied risk-adjusted discount rate that makes the DNPV equal to zero are equal to 23% (i.e., $3\% + 20\%$) and 33% (i.e., $3\% + 20\% \times 1.5$) for λ equal to 0 and 1, respectively.

⁸ See Equation (A.3d) in Appendix A.

⁹ For a synthetic insurance, this factor can be viewed as a safety factor against an error in the estimated expected value of the loss.

Table 2
Cash flows analysis.

Period	Investment	Revenues	Value of the project (Eq. (A.7))	Synthetic insurance premium
0	$\tilde{M}_0$	–	$\tilde{M}_0$	–
1	–	$\tilde{V}$	$\tilde{M}_1$	$R_1 = \rho \times (1 + \lambda c_v) \tilde{M}_0$
2	–	$\tilde{V}$	$\tilde{M}_2$	$R_2 = \rho \times (1 + \lambda c_v) \tilde{M}_1$
⋮	–	⋮	⋮	⋮
n	–	$\tilde{V}$	$\tilde{M}_n$	$R_n = \rho \times (1 + \lambda c_v) \tilde{M}_{n-1}$

6.3. Case 2: Loss of Revenues due to a Temporary Shutdown

For this particular case, it is assumed that there is a probability equal to ρ that the project may not generate any revenues temporarily and that the probability of such an event is the same throughout the contract period. As illustrated below, at any time during the contract period, there are two possible events: either the project shuts down temporarily for one period or else it continues uninterrupted to generate revenues. Table 3 illustrates this binary probability distribution, where the expected value of the revenue, $\tilde{V}$, is equal to $V \times (1 - \rho)$.

Using the project risk profile in Table 3, it follows that the expected value of the loss at any period of time ($\tilde{L}_i = \rho \times (1 - \rho) \times V$) with a standard deviation ($s_v = \sqrt{\rho \times (1 - \rho)} V$) and a coefficient of variation ($c_v = \sqrt{\rho / (1 - \rho)}$) can be estimated from this table (3rd Column). Thus, again assuming that the standard deviation principle is an appropriate measure of the synthetic insurance premium, the synthetic insurance premium (R_i) can be written as:

$$R_i = \tilde{L}_i(1 + \lambda c_v) = \Theta \tilde{V} \quad (21)$$

For this example, as in the previous case, a cash flow model can be developed similar to that shown in Table 2, with the exception that the synthetic insurance premium is constant with time and equal to Eq. (21). Because the risk associated with not getting revenues temporarily is considered by the synthetic insurance premium, Eq. (15) can be used as in the previous case to estimate the value of the investment project. For this case, the implied equivalent risk adjusted discount rate to account for the probability of temporary project shutdown at any time during the contract period is given by:

$$F(\alpha) = F(r)(1 - \Theta) \quad (22)$$

Eq. (22) must be solved¹⁰ iteratively to find a value for α . It follows from Eq. (21) that if $c_v = 0$ (i.e., no uncertainty), then the discount is equal to the risk free rate. Also, it follows directly from Eq. (21) that the risk factor for this case is constant and equal to Θ .

Using Eq. (22) with the same parameters as Case 1 (i.e., $\rho = 20\%$, $r = 3\%$, and $n = 10$), the appropriate implied risk-adjusted discount rate that makes the DNPV equal to zero for $\lambda = 0$ (i.e., risk neutral), was found iteratively to be equal to 7.6% (Case 2a). If a loading factor $\lambda = 1$ were used, the implied risk adjusted

Table 3
Project risk profile in any period (temporary shutdown).

Event (E)	Probability, $\Pr(E)$	Value of the revenue	Insurance payout
Shutdown	ρ	0	$\tilde{V} = V \times (1 - \rho)$
No shutdown	$1 - \rho$	V	0

discount rate would be 10.7% (Case 2b). Note that for Case 2, different to Case 1, the appropriate implied risk-adjusted discount rate that is consistent with the project risk profile depends upon the duration of the project (i.e., the number of periods, n). Thus, for constant risk profile, different risk adjusted discount rates would be appropriate depending on the project duration. For instance, for the same case as above but with $n = 5$, the appropriate implied risk adjusted discount rate would be 11.35% for $\lambda = 0$ and 16.9% for $\lambda = 1$.

Rather than estimating a synthetic insurance premium based on a binary probability distribution function, no-arbitrage principles used in the financial markets to value derivatives can be used to estimate the synthetic insurance premium. To this end, the uncertainty regarding the expected revenues is correlated to the expected remaining total annual revenues (P), assuming P follows the stochastic process proposed by Pindyck (1993) for projects that take time to build. The resulting risk premium for such a case can be obtained from the linearized expressions Eqs. (23) and (24)¹¹ derived by Espinoza (2011):

$$R_i = \Omega \tilde{V} \quad (23)$$

where:

$$\Omega = \left(N\left(\frac{v}{2\sqrt{2}}\right) - N\left(-\frac{v}{2\sqrt{2}}\right) \right) \quad (24)$$

As in the previous case, the risk factor is constant and equal to Ω . The stochastic volatility parameter (v) in Eq. (24) can be estimated as a function of the time independent coefficient of variation c_v (Pindyck, 1993) calculated from the stationary date at $t = 0$.

$$v = \left(\frac{2c_v^2}{1 + c_v^2} \right)^{1/2} \quad (25)$$

Note that the risk premium given by Eq. (23) is similar to Eq. (21). Hence, by substituting Ω instead of Θ , Eq. (22) can be used to find the value for α (the implied risk adjusted discount rate). It follows from Eq. (25) that if $c_v = 0$ (i.e., no uncertainty), then the synthetic insurance premium is zero and thus α is equal to the risk free rate. Using the same parameters as in the previous example (i.e., $\rho = 20\%$; $r = 3\%$, and $n = 10$), the value of α that makes the DNPV equal to zero would be 6.25%, which is slightly less than the binomial case with $\lambda = 0$ (i.e., risk neutral).

¹⁰ Eq. (22) can be straightforwardly derived by replacing Eq. (A.5) in Appendix B with Eq. (21), and setting the resulting equation equal to Eq. (A.6).

¹¹ A detailed derivation of Equation (24) is presented in Appendix B.

6.4. Case 3: Loss of revenues due to a change in market prices

This case assumes that revenues are from the sale of a commodity (e.g., oil, electricity, copper) and the cost of the synthetic insurance risk premium is due to market price fluctuations. For discussion purposes, it is assumed the commodity is a traded security with an implied volatility of σ and prices that follow a Brownian motion stochastic process. Based on no-arbitrage principles, despite an asset's actual expected growth rate, the future price at any given year ($\tilde{V}_i$) is expected to grow at a rate $(r - \delta)$ where δ represents the dividends paid by the asset (or the convenience yield in the case of commodities):

$$\tilde{V}_i = \tilde{V}_0(1 + r - \delta)^i \quad (26)$$

Because the uncertainty regarding future revenues from the sale of a traded commodity that evolves stochastically with time and follows a Brownian motion process, the closed form solutions for put options obtained based on risk neutral considerations by Black and Scholes (1973) can be used to price a synthetic insurance premium for the expected cash flows given by Eq. (26). Hence, the present value of the annual synthetic insurance premium (R_i^o) is given by:

$$R_i^o = \Omega_i^o \tilde{V}_0 \quad (27)$$

$$\Omega_i^o = e^{-\delta i} \left[-N(-0.5\sigma\sqrt{i}) + N(0.5\sigma\sqrt{i}) \right] \quad (28)$$

Eq. (28) represents the risk factor for the project and the superscript “o” is used to indicate that it corresponds to its present value. Note that for this particular example, because the revenues are correlated with a traded commodity, investors could actually hedge against this risk and obtain a financial product (i.e., put options) to protect against a drop in the expected value of the asset (at least for a project of short-term duration). Like in the previous cases, a non-linear equation relates the equivalent risk-adjusted discount rate to the risk free rate (see Appendix C). As shown in the appendix, it follows that if $\sigma = 0$ (i.e., no market uncertainty), then the synthetic insurance premium is zero and thus the implied risk-adjusted discount rate is equal to the risk free rate.

This example shows how the proposed DNPV methodology can be used to estimate the synthetic insurance premiums for market (non-idiosyncratic) risks that is consistent with pricing of idiosyncratic risks. This illustrates that, even for cases where only market risk is present, rather than selecting a discount rate (e.g., using CAPM) to discount future cash flows from a project, existing derivatives can be used to estimate the price of a synthetic insurance premium to estimate the project DNPV. As in the previous cases, in line with the CE method, future uncertain cash flows were reduced to create a stream of synthetically riskless cash flows. However, for this case, the reduction was estimated based upon no-arbitrage concepts rather than utility theory concepts.

6.5. Implied risk adjusted discount rate

A comparison of the implied risk-adjusted discount rates for the first two cases, where the risk associated with the project revenues is due to technical considerations (e.g., plant shutdown), is presented in Fig. 3 for contract periods that vary from 1 to 20 years. For simplicity, without loss of generality, the risk free rate was assumed constant for the different contract durations. As expected, the equivalent risk-adjusted discount rate profile for projects subject to risk of a permanent shutdown are higher than those subject to risk of a temporary shutdown. For risk of permanent shutdown (Case 1), constant risk premiums (i.e., the interest rate above the risk-free rate) of 20% and 28% (for λ equal to 0 and 1, respectively) were estimated for the contract period analyzed. For the case of temporary shutdown (Case 2), the risk premiums monotonically decrease with time, converging for $t \gg 20$ years to 3.76%, 4.18%, and 3.52%, respectively. As shown in Fig. 3, except for the permanent shutdown cases, a constant risk premium could not simply be added to the risk free rate and applied to discount future revenues without considering the contract period.

Cases 1 and 2 can be related to infrastructure projects with an uncertain stream of cash flows due to technical risks (e.g., amount of radiation for solar energy plants; amount of wind for wind energy plants; volume of water for hydropower plants). For these cases, if tariffs have been pre-negotiated, the revenue risk would be mostly attributed to having less radiation (or wind, or water) than expected. The risk of higher expenditures for operations and maintenance can be estimated based on operational experience and/or historical information from similar plants. These cases show that the practice of selecting a constant discount rate to estimate a future stream of cash flow independent of the contract period is not appropriate. Both cases can also be related to the lending business where initial funding is provided for a stream of future cash flows. In such case, the probability of permanent shutdown (Case 1) can be related to the probability of loan default whereas the probability of temporary shutdown (Case 2) can be

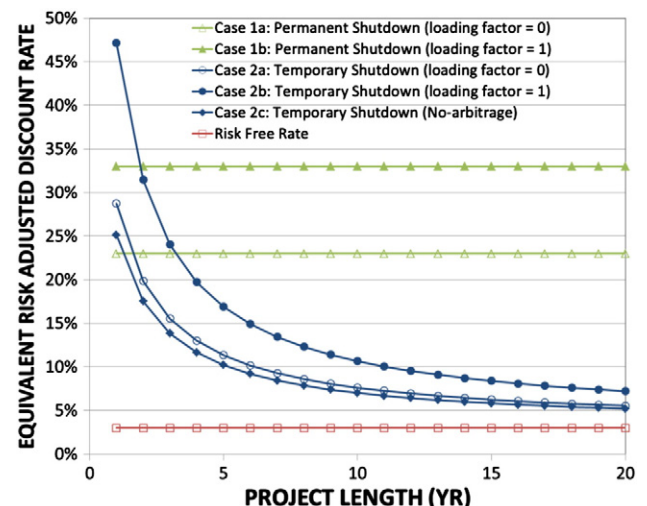


Fig. 3. Comparison of risk profiles.

related to the probability of the borrower becoming delinquent on some installments.

Fig. 4 shows the implied risk adjusted discount rates for the case of market risk only for four different volatilities and for contract periods that vary from 1 to 20 years. As shown in this figure, as in the case of technical risk, assuming that constant volatility over the period analyzed, the implied risk-adjusted discount rate decreases monotonically with the length of the project. Thus, as in Case 2, a constant discount rate cannot be applied to projects of different durations.

The risk factor (and CE factor which is simply one minus the risk factor) as a function of time are presented in Fig. 5 for the three cases analyzed. As shown in this figure, the risk factor (open symbols in Fig. 5) varies with time and can be monotonically increasing (e.g., market risk), constant (e.g., risk of temporary shutdown), or monotonically decreasing with time (e.g., permanent shutdown). For comparative purposes, the corresponding CE factors for the cases described above are also presented in Fig. 5. It is interesting to note that, for the cases analyzed, despite the shape of the risk factors (or CE factors), the implied risk-adjusted discount rate always decrease with time (Figs. 3 and 4).

Except for the permanent shutdown (Case 1), the cases analyzed illustrate that the use of a single risk-adjusted discount rate to value projects with different contract durations is not appropriate. The cases analyzed also indicate that the implied risk-adjusted discount rate should decrease for longer contract periods. As shown in the application example below, the different risk factors evaluated above can be easily combined to evaluate projects that are affected by various risk profiles. Similar ideas can be applied to more complex cash flows where risk profiles can be assigned to the revenues and/or expenditures to estimate the project DNPV.

7. Case study

In what follows, the advantages of the proposed DNPV method are illustrated using a hypothetical but realistic case study. The cash flows are evaluated first using the classical NPV method, then using the ROV method, and followed by an application of the proposed DNPV. A business executive is

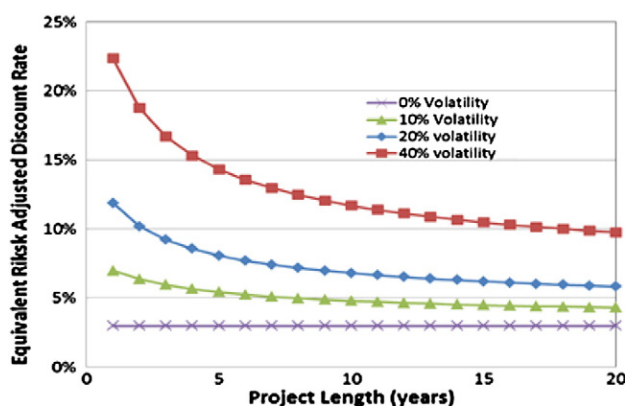


Fig. 4. Comparison of market risks.

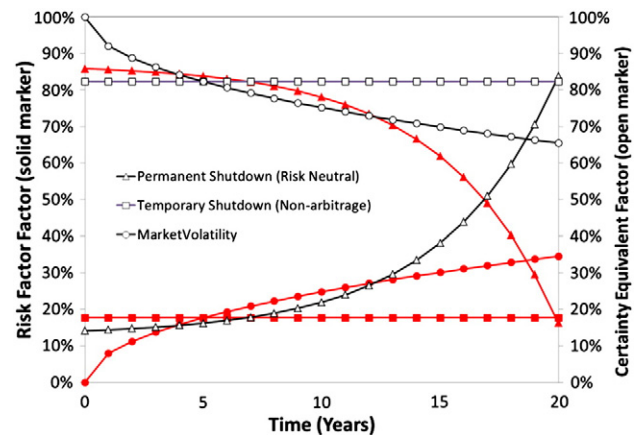


Fig. 5. Evolution of risk factors (open)/certainty equivalent factors (filled) with time.

considering an initial 510 million capital investment (Phase 1) to extract a commodity for a period of ten years. Based on current prices, the expected annual revenues is 874 million with operating expenses (OPEX) equal to 90% of the generated revenues for a period of ten years. The business executive has also the option to expand the project to double its capacity and reduce its OPEX of the additional production to 80% of the additional revenues starting in year 4 by investing an additional 550 million in year 3 (Phase 2). Because of the price volatility of the commodity, the business executive considers that a WACC of 21% is appropriate for this project (the risk free rate r is 3%). The nominal cash flows along with the estimated NPV for the entire project (Phases 1 and 2) are shown in Table 4a (for simplicity, inflation is assumed to be the same as the risk free rate). Because the project NPV is negative, the project is not considered feasible. Advocates of ROV suggest that because project managers are not committed to make the second phase of the investment, the NPV method in Table 4a ignores the value of the flexibility of the project (e.g., Yeo and Qiu, 2003) which could be significant given the high annual volatility of the revenues ($\sigma = 60\%$). To investigate this additional value, the cash flows are divided into Phase 1 (NPV = $-\$109.9$ million) and Phase 2 (NPV = $\$106.6$ million) and are summarized in Tables 4b and 4c, respectively. As expected, the sum of these two valuations yields the NPV calculated in Table 4a.

Now, assuming that cash flows from the expansion project from year 4 to 10 follow a Brownian motion stochastic process, the value of the call option (the supposed value of the manager's flexibility) can be estimated using the Black and Scholes closed form solution for a call option. A typical ROV analysis would set the value of the asset (S) as the NPV (using WACC = 21%) of the cash flows from Table 4c, that is $S = \$417.1$ million with an exercise price $X = \$550$ million. Thus, using $r = 3\%$ and $\sigma = 60\%$, the value of a 3-year call option is $\$142.2$ million. Thus, using the ROV methodology, the value of this investment would now be $\$32.4$ million ($-\$109.9 + \142.2), in other words, because of the high volatility, the value of the manager's flexibility to execute the expansion has added $\$35.7$ million of value making the total NPV positive, therefore the project is deemed feasible.

Table 4a

Cash flow distribution for entire project (NPV = −3.3 million).

Year	0	1	2	3	4	5	6	7	8	9	10
Revenues		900.0	927.0	954.8	1966.9	2025.9	2086.7	2149.3	2213.8	2280.2	2348.6
OPEX		−810.0	−834.3	−859.3	−1672	−1722	−1772	−1827	−1882	−1938.2	−1996.3
CAPEX	−510	0.0	0.0	−550	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cash flow	−510.0	90.0	92.7	−454.5	295.0	303.9	313.0	322.4	332.1	342.0	352.3

As discussed above, the main problem with the application of NPV is the arbitrary selection of the discount rate to calculate the value of the project and its inability to account for manager's flexibility to alter the course as events unfolds and new information arrives. While in principle, ROV takes care of the latter, the added complexity does not result in a better valuation due to several drawbacks: (i) the option value is used to complement NPV thus inheriting the flaws of this method; (ii) some of the parameters needed to price the option are estimated using the same arbitrary discount rate; (iii) it assumes that the NPV of the project without expansion remains unchanged whether or not the option is exercised; (iv) the analysis implies that the option is free and immediately exercisable giving the false impression that the option value is always greater than zero.

In the example above, using the arbitrary discount rate of 21%, the project NPV for Phase 1 is very negative (−\$109.9 million). The option to expand in year 4 (by investing 550 million in year 3) supposedly has a value of 142.2 million, thus making the total NPV 32.4. However, the option to expand is exercised only if the expected cash flows are greater than 550 million. If the option is not exercised (i.e., the expansion not constructed), it is likely that during the first three years the revenues were lower and/or cost higher than expected thus making the NPV of the non-expanded project even lower than −109.9 million in the case of not exercising the option. Even if the revenues were as the manager forecasted in Phase 1, at the time of the second CAPEX is made, because it takes time to build the expansion (one year in this example), during the investment period (or soon after), market conditions during Phase 2 could change resulting in a negative NPV of the expansion.

The same project is analyzed below using the proposed DNPV to illustrate its application and assuming that the project cash flows are affected by four risks, namely: market risk, temporary shutdown risk, permanent shutdown risk and environmental risk. The market risk is described by the 60% volatility used for the ROV method whereas it is assumed that the business managers have industry information that allows them to assume a 2% chance of temporary shutdowns at any time and a small likelihood

(0.01%) of complete shutdown which is also coupled to a \$2 billion of environmental liability. The cost of risk for the first three identified risks can be estimated using Eqs. (27), (21), and (14), respectively, whereas the cost of risk per year for the environmental liability can be estimated simply as $0.01\% \times \$2 \text{ billion} = \0.2 million .

Hence, using the proposed DNPV methodology, the project cash flows calculated in Tables 4b and 4c can be de-risked as shown in Tables 5a and 5b, respectively. The resulting DNPV for Phases 1 and 2 are −\$12.3 million and \$138.2 million, for a total value of \$126 million. The implied risk adjusted discount rates for Phases 1 and 2 are 15.2% and 19.7%, respectively (and 17% for the entire Phase 1 and 2). Although implied risk-adjusted discount rates have been calculated, this is done mainly to relate to a parameter that is very much in use by the investment community.

The present value for the total cost of risk for Phases 1 and 2 are \$376 and \$582 million, respectively. As described in Section 2, DNPV can be compared to a call option and the total value of the project is simply the sum of the DNPV for each phase. In theory, an investment should be made if $\text{DNPV} > 0$ (i.e., it is in the money by at least the cost of project risk). In this example, although the value of the project for the initial phase is below zero, this negative value only represents 3.3% of the total cost of risk. In addition, there is a significant future upside if Phase 1 is developed as demonstrated by the value of the DNPV of the expansion.

In the analysis presented in Tables 4a, 4b, 4c, 5a and 5b, it is implicitly assumed that revenues and costs are strongly correlated (i.e., if the revenues goes up, OPEX also goes up in tandem and vice versa) so in calculating the cost of risk only the difference between revenues minus OPEX need to be considered. This apparent innocuous assumption can be disastrous for the investor if this is not the case. For instance, if the estimated OPEX in Tables 4b and 4c are relatively stable and are not expected to deviate significantly from the values shown in Tables 4b and 4c (which it may be viewed as a project good characteristic as there is cost certainty), then the cost for market risk would corresponds to a put option for the revenues only which are ten and five times

Table 4b

Cash flow distribution without expansion (NPV = −109.9 million).

Year	0	1	2	3	4	5	6	7	8	9	10
Revenues		900.0	927.0	954.8	983.5	1013.0	1043.3	1074.6	1106.9	1140.1	1174.3
OPEX		−810.0	−834.3	−859.3	−885.1	−911.7	−939.0	−967.2	−996.2	−1026.1	−1056.9
CAPEX	−510.0										
Cash flow	−510.0	90.0	92.7	95.5	98.3	101.3	104.3	107.5	110.7	114.0	117.4

Table 4c

Cash flow distribution of the expansion (NPV = 106.6 million).

Year	0	1	2	3	4	5	6	7	8	9	10
Revenues		0.0	0.0	0.0	983.5	1013.0	1043.3	1074.6	1106.9	1140.1	1174.3
OPEX		0.0	0.0	0.0	−786.8	−810.4	−834.7	−859.7	−885.5	−912.1	−939.4
CAPEX				−550.0							
Cash flow	0.0	0.0	0.0	−550.0	196.7	202.6	208.7	214.9	221.4	228.0	234.9

greater than the cash flows for Phases 1 and 2, respectively. As a result the costs of risk are increased ten-fold for Phases 1 and five-fold for Phase 2 making the calculated DNPV equal to −\$3.1 billion and −\$2.1 billion, respectively for a total project value of −\$5.2 billion. This clearly indicates that although a 60% volatility of the revenues (which are in the order of \$1 billion per year) can result in upside value (about the same as the estimated cost of risk, that is, \$2.1 billion), the flip side is that the same project volatility can lead to significant losses if the market were to turn negative after the initial investment for the expansion is made.

This hypothetical example shows the power of the proposed DNPV and its ability to de-risk the expected cash flows in a consistent and familiar manner. The methodology empowers business executives to identify the project potential sources of risk, price those risks, and understand how they could affect the value of the project. As a result, business executives can make informed/better management decisions that can lead to an improved value of the project and the business bottom line.

8. Conclusions

Combining concepts from insurance and capital budgeting, the DNPV valuation method, which is an extension of the certainty equivalent method and equivalent to real options, has been presented. Using this method, risk is decoupled from the time value of money and is accounted for in the numerator as a synthetic insurance premium allowing for consistent and transparent risk pricing. The concept of synthetic insurance is introduced within the realm of project valuation to price the risk associated with obtaining lower revenues or higher costs than originally expected and these synthetic insurance products are treated as “real” costs to the project which are then subtracted from the expected cash flows to de-risk the estimated cash flows. Because the risks associated with the project are accounted for by these synthetic insurance

products, the riskless cash flows can be discounted using a risk-free rate. It is shown that DNPV is equivalent to the profit–loss profile of a long call option position or a protective put.

Several simplified examples with a single risk source for the revenue stream were presented to illustrate the application of the DNPV methodology and to develop the building blocks for a more realistic analysis. The simplified examples demonstrate that once the risks of the cash flow streams have been identified, riskless cash flows can be easily estimated. Because risk is treated as a cost (i.e., an insurance premium), the calculated cost of risk can be easily implemented in a traditional capital budgeting application as it was demonstrated in Section 7. More importantly, partial results (i.e., cost of risk) and final values (i.e., DNPV) can be easily explained to decision makers. The proposed DNPV methodology can be easily extended to account for the different risk profiles affecting each of the sources of revenues and/or expenditures without resorting to decision makers’ gut instincts to select an appropriate discount rate for the project.

The introduction of probability and stochastic methods to price the cost of risk (i.e., synthetic insurance premiums) provides DNPV users with a direct link between traditional risk management and financial performance. As demonstrated in the cases analyzed and the example, the discount rates were estimated based on the project risk profile rather than assumed heuristically as is typically the case. This constitutes a significant step forward in the area of financial risk management as any steps taken to mitigate a project’s potential downside can have a direct and quantifiable impact on the project financial performance. The proposed DNPV method will also allow managers of large corporations to mine available data using business analytic techniques to develop risk profiles of their revenues/expenditures that can then be used to estimate synthetic insurance premiums, thus improving their valuation results.

Although this procedure was initially developed to evaluate long-term infrastructure projects (e.g., solar energy plants, wind

Table 5a

Decoupled cash flow distribution without expansion (DNPV = −12.3 million).

Year	0	1	2	3	4	5	6	7	8	9	10
Cash flows	−510	90.0	92.7	95.5	98.3	101.3	104.3	107.5	110.7	114.0	117.4
Market risk		−20.6	−28.7	−34.6	−39.3	−43.3	−46.8	−49.8	−52.5	−54.9	−57.1
Temporary shutdown risk		−1.76	−1.82	−1.87	−1.93	−1.99	−2.04	−2.11	−2.17	−2.23	−2.30
Permanent shutdown risk		0.09	0.08	0.07	0.07	0.06	0.05	0.04	0.03	0.02	0.01
Environmental liability		0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Decoupled cash flow		67.4	61.9	58.8	56.8	55.7	55.2	55.3	55.8	56.6	57.8

Table 5b

Decoupled cash flow distribution without expansion (DNPV = \$138.2 million).

Year	0	1	2	3	4	5	6	7	8	9	10
Cash flows		0.0	0.0	−550	196.7	202.6	208.7	214.9	221.4	228.0	234.9
Market risk	0.0	0.0	0.0	0.0	−78.7	−86.7	−93.6	−99.7	−105.0	−109.9	−114.2
Temporary shutdown risk	0.0	0.0	0.0	0.0	−3.86	−3.97	−4.09	−4.21	−4.34	−4.47	−4.60
Permanent shutdown risk		0.0	0.0	0.0	0.13	0.12	0.10	0.08	0.06	0.04	0.02
Environmental liability		0.00	0.00	0.00	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Decoupled cash flow		0.0	0.0	−550	113.8	111.6	110.7	110.8	111.7	113.4	115.8

energy plants, nuclear power plants, hydropower plant), the proposed methodology can be applied to all other industries (e.g., pharmaceutical, oil, transportation, utilities, retail, insurance). Future research can be devoted to collect, compile and process relevant data to develop industry-wide PDFs such that better risk evaluation can be performed.

Appendix A

Following the definition of DNPV, the expected values of the constant stream of revenues $\tilde{V}$ over n periods can be used to estimate the expected value of the investment ($\tilde{M}_o$) as:

$$\begin{aligned} DNPV &= \sum_{i=1}^n \frac{(\tilde{V}-R_i)}{(1+r)^i} - \tilde{M}_o \\ &= \sum_{i=1}^n \frac{\tilde{V}}{(1+r)^i} - \sum_{i=1}^n \frac{R_i}{(1+r)^i} - \tilde{M}_o = 0 \end{aligned} \quad (A.1)$$

where R_i is the synthetic insurance premium. For comparative purposes, the value of the investment opportunity is also calculated using the traditional NPV method and discounting the future revenues by a risk-adjusted discount rate as:

$$NPV = \sum_{i=1}^n \frac{\tilde{V}}{(1+\alpha)^i} - \tilde{M}_o = 0 \quad (A.2)$$

Using the summation series identity below:

$$\sum_{i=1}^n a^i = \frac{a(1-a^n)}{(1-a)} \quad (A.3)$$

and substituting $a = (1+x)^{-1}$ in Eq. (A.3), the following expression is obtained:

$$F(x) = \sum_{i=1}^n \frac{1}{(1+x)^i} = \frac{(1-(1+x)^{-n})}{x} \quad (A.4)$$

Then, since $\tilde{V}$ is constant, the stream of revenues can be removed from the summation in Eqs. (A.1), (A.2) and (A.4) can be used to simplify both equations as:

$$\tilde{M}_o = \tilde{V} F(r) - \sum_{i=1}^n \frac{R_i}{(1+r)^i} \quad (A.5)$$

$$\tilde{M}_o = \tilde{V} F(\alpha) \quad (A.6)$$

For the case of a complete shutdown, the synthetic insurance premium R_i is associated with the expected loss of the remaining future revenues ($\tilde{M}_i$), which can therefore be estimated using the equivalent risk-adjusted discount rate as the value of the project at period i as:

$$\tilde{M}_i = \frac{\tilde{V}}{\alpha} - (1+\alpha)^{i-1} \left[\frac{\tilde{V}}{\alpha} - \tilde{M}_o \right] \quad (A.7)$$

Hence, the value of the project can be estimated if the equivalent risk adjusted discount rate (α) is calculated. If the value of the synthetic insurance is described by the standard deviation principle as:

$$R_i = \Theta \tilde{M}_i \quad (A.8)$$

where:

$$\Theta = \rho \times (1 + \lambda c_v) \quad (A.9)$$

Setting Eq. (A.5) equal to (A.6) and replacing Eqs. (A.7) and (A.8) in the resulting equation yields:

$$\tilde{V} F(r) - \sum_{i=1}^n \left(\Theta \left\{ \frac{\tilde{V}}{\alpha} - (1+\alpha)^{i-1} \left[\frac{\tilde{V}}{\alpha} - \tilde{V} F(\alpha) \right] \right\} \right) \frac{1}{(1+r)^i} = \tilde{V} F(\alpha) \quad (A.10)$$

Because most of the expression inside the summation is constant with respect to i , it follows from Eq. (A.10) that:

$$F(r) - \Theta \frac{F(r)}{\alpha} + \frac{\Theta}{1+\alpha} \left[\frac{1}{\alpha} - F(\alpha) \right] \sum_{i=1}^n \frac{(1+\alpha)^i}{(1+r)^i} = F(\alpha) \quad (\text{A.11})$$

Using Eq. (A.3) with $a = (1+\alpha)(1+r)^{-1}$ in Eq. (A.11) yields:

$$\sum_{i=1}^n \left(\frac{1+\alpha}{1+r} \right)^i = \frac{1+\alpha}{r-\alpha} \left[1 - \left(\frac{1+\alpha}{1+r} \right)^n \right] \quad (\text{A.12})$$

Using Eq. (A.4) to define $F(\alpha)$ and $F(r)$ and simplify Eq. (A.12), and replacing the resulting equation into Eq. (A.11), results in:

$$F(r) \left(1 - \frac{\Theta}{\alpha} \right) + \frac{\Theta}{r-\alpha} \left[\frac{1}{\alpha} - F(\alpha) \right] \left[\frac{rF(r) - \alpha F(\alpha)}{1 - \alpha F(\alpha)} \right] = F(\alpha) \quad (\text{A.13})$$

which can be further simplified to:

$$F(r) \left(1 - \frac{\Theta}{\alpha} \right) + \frac{\Theta}{(r-\alpha)\alpha} [rF(r) - \alpha F(\alpha)] = F(\alpha) \quad (\text{A.14})$$

or:

$$\frac{F(r)}{F(\alpha)} \left(1 - \frac{\Theta}{\alpha} + \frac{\Theta}{r-\alpha} \frac{r}{\alpha} \right) - \left[\frac{\Theta}{r-\alpha} + 1 \right] = 0 \quad (\text{A.15})$$

Eq. (A.15) is a nonlinear equation as a function of α . A trivial solution to Eq. (A.15) can be found by setting the quantities inside the brackets equal to zero and solving the resulting system of equations. Then it follows that:

$$\alpha = r + \Theta \quad (\text{A.16})$$

Finally, the value of the project can be expressed from Eqs. (A.6) and (A.16) as:

$$\tilde{M}_o = \tilde{V} F(r + \Theta) \quad (\text{A.17})$$

Since $\tilde{M}_o = (1-\rho)M_o$ and $\tilde{V}_o = (1-\rho)M_o$, it follows that:

$$M_o = V F(r + \Theta) \quad (\text{A.18})$$

Also, defining a risk factor (η) as the ratio of the synthetic insurance premium over the revenues, then from Eqs. (A.7), (A.8), (A.9), and (A.16), it follows that:

$$\eta_i = \left(\frac{1}{r+\Theta} - (1+r+\Theta)^{i-1} \left[\frac{1}{r+\Theta} - F(r+\Theta) \right] \right) \Theta \quad (\text{A.19})$$

Appendix B

The uncertainty regarding the expected revenues is assumed to be correlated to the expected remaining total payment ($\tilde{P}$) in a given year, with $\tilde{P}$ following the same stochastic process given by Pindyck (1993) proposed for projects that takes time to build.

$$dP = -Idt + v(IP)^{1/2} dz \quad (\text{B.1})$$

where dz is the increment of uncorrelated Wiener process (i.e., it is normally distributed with a mean value of zero and a standard deviation equal to $\sqrt{dt}$), I is the expected revenue rate, and v is the stochastic volatility parameter of the revenue. The second term on the right hand side of Eq. (B.1) represents the uncertainty of the cash flows. The equation indicates that every time that revenue is obtained, the uncertainty regarding future payment is reduced as P is reduced (i.e., the second term gets smaller). For the problem analyzed, the expected revenue rate can be estimated as:

$$I = \frac{n\tilde{V}}{T} \quad (\text{B.2})$$

where T is the contract period, $\tilde{V}$ is the constant stream of revenues to obtain each period, and n is the number of periods. Because Eq. (B.1) leads to a non-linear partial differential equation, its solution requires numerical algorithms (Pindyck, 1993). However, an approximate linear equivalent expression for Eq. (B.1) was derived by Espinoza (2011) by expressing P in terms of the total revenue Q as:

$$Q = P + It \quad (\text{B.3})$$

Assuming that Q represents the total annual revenue, Eq. (B.1) can be approximated by the following equivalent linear differential equation:

$$\frac{dQ}{Q} = d(\ln Q) = \sigma \varepsilon \sqrt{dt} \quad (\text{B.4})$$

$$\sigma = \frac{v}{\sqrt{2}} \quad (\text{B.5})$$

where ε is a normally distributed random variable with mean of zero and a standard deviation of 1.0, and σ is the equivalent linear stochastic volatility. Eq. (B.4) is the equivalent linear representation of the more accurate non-linear stochastic process given by Eq. (B.1). Also, the stochastic process given by the linearized Eq. (B.4) is similar to the well-known stochastic process used for modeling stock options. As Q evolves stochastically with time, the expected loss at the end of a year (Q is assumed to represent the total annual revenue) is given by the boundary condition:

$$R_t = \text{Max}(X - Q, 0) \quad (\text{B.6})$$

Thus, the solution of Eq. (B.4) subject to the boundary condition given by Eq. (B.6) results in the well-known closed form solution for European Put options (i.e., the right to sell an asset at a pre-specified price) evaluated at the beginning of the period (Black and Scholes, 1973):

$$R_i^{t-1} = -Qe^{-\delta t}N(-d_t) + Xe^{-rt}N(\sigma-d_t) \quad (\text{B.7})$$

where:

$$d_t = \frac{\ln(Q/X) + (r-\delta + \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}} \quad (\text{B.8})$$

and δ is a convenience yield. For this particular case, $\delta = r$ (Espinoza, 2011) and $t = 1$ year. Eq. (B.7) is the value of an European put option which gives the right to the option holder to sell the asset Q at a pre-specified price X at a specified date (in this case, at the end of the year) by paying a premium R_i^{t-1} at the beginning of any given year. This is basically an insurance policy that would be exercised if the value of the asset Q dropped below X at the end of each year. Since Q represents the average value of the total payments for one year, then, to protect this expected value, the pre-specified exercise price at each time period must be set equal to $X = Q = \tilde{V}$. Hence, Eq. (B.7) reduces to:

$$R_i^{t-1} = \tilde{V}e^{-r}\left(-N\left(-\frac{v}{2\sqrt{2}}\right) + N\left(\frac{v}{2\sqrt{2}}\right)\right) \quad (\text{B.9})$$

Because the closed form solution in Eq. (B.9) gives the value of the synthetic insurance premiums at the beginning of a period, the equation needs to be multiplied by e^{-r} to be at the same time as the revenue. Hence, the risk premium for each year can be expressed as:

$$R_i = \Omega\tilde{V} \quad (\text{B.10})$$

where:

$$\Omega = \left(N\left(\frac{v}{2\sqrt{2}}\right) - N\left(-\frac{v}{2\sqrt{2}}\right)\right) \quad (\text{B.11})$$

Appendix C

For this case, it is assumed that the annual revenues have the same expected market value (i.e., quoted prices) of $\tilde{V}_o$ and there is a perfectly correlated security in the market that can be used to price the project risk and the risk can be modeled using the Black and Scholes (1973) model. Thus, the future price at any given year can be estimated as:

$$\tilde{V}_i = \tilde{V}_o(1+r)^i \quad (\text{C.1})$$

The corresponding synthetic insurance can be estimated using Eq. (B.7) and replacing the value of the asset (Q), the

exercise price (X), and the duration of the option (t) by $\tilde{V}_o$, $\tilde{V}_i$, and i , respectively:

$$\frac{R_i}{(1+r)^i} = -\tilde{V}_oe^{-\delta i}N(-d_i) + \tilde{V}_ie^{-ri}N(\sigma-d_i) \quad (\text{C.2})$$

Approximating $\tilde{V}_i$ in Eq. (C.1) as $\tilde{V}_oe^{(r-\delta)i}$, replacing it in Eq. (C.2), and then substituting the resulting equation in Eq. (A.1) yields:

$$\tilde{M}_o = n\tilde{V}_o - \tilde{V}_oe^{-\delta i}\sum_{i=1}^n\Omega_i^o \quad (\text{C.3})$$

$$\Omega_i^o = -N\left(-\frac{1}{2}\sigma\sqrt{t}\right) + N\left(\frac{1}{2}\sigma\sqrt{t}\right) \quad (\text{C.4})$$

As in the previous cases, for the purposes of comparison, the value of the project is also estimated using an implied risk-adjusted discount rate as:

$$NPV = \sum_{i=1}^n \frac{\tilde{V}_o(1+r)^i}{(1+\alpha)^i} - \tilde{M}_o = 0 \quad (\text{C.5})$$

Using Eq. (A.3) with $a = (1+\alpha)^{-1}(1+r)$ in Eq. (C.5) and using Eq. (A.4) to define $F(\alpha)$ and $F(r)$ yields:

$$\sum_{i=1}^n \left(\frac{1+r}{1+\alpha}\right)^i = \frac{1+r}{\alpha-r} \left[1 - \left(\frac{1+r}{1+\alpha}\right)^n\right] = \frac{1+r}{\alpha-r} \left[\frac{\alpha F(\alpha) - r F(r)}{1 - r F(r)}\right] \quad (\text{C.6})$$

Replacing Eq. (C.6) into Eq. (C.5) and setting the resulting expression equal to Eq. (C.3) yields:

$$\frac{1+r}{\alpha-r} \left[\frac{\alpha F(\alpha) - r F(r)}{1 - r F(r)}\right] = \left(n - e^{-\delta i} \sum_{i=1}^n \Omega_i^o\right) \quad (\text{C.7})$$

which can be further simplified to the following recursive equation:

$$\alpha = r \frac{H(r) - F(r)}{H(r) - F(\alpha)} \quad (\text{C.8})$$

where:

$$H(r) = \left(n - e^{-\delta i} \sum_{i=1}^n \Omega_i^o\right) \frac{(1 - r F(r))}{1 + r} \quad (\text{C.9})$$

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